

Compressor Q

One compressor with four voicings, plus a four-band EQ that can go before or after it.



Overview

Compressor Q is a single compressor that can take on four characters. VCA is a punchy loudness-sensing (RMS) compressor whose ratio goes past infinity into negative ratios. BUS is a smooth, peak-sensing glue compressor that always works on both sides together. RIDE is a transparent leveller with a wide knee. OPTO behaves like an optical cell: a soft attack and a release that lets go quickly at first, then slowly.

The SIDE-CHAIN section either stops the lows from driving the compressor (SC HPF) or compresses only the lows (SPLIT). MIX blends in the dry signal for parallel compression.

Below the compressor sits a four-band EQ with low and high cut filters. EQ FIRST puts it before the compressor, EQ AFTER after it; the unit the signal meets first is shown on top.

Reach for it for

- Vocals that need levelling and tone in one place (RIDE or OPTO)
- Drum smashing and parallel compression (VCA with MIX)
- Gentle glue on a mix or group bus (BUS)
- Controlling only the low end of a bass or a mix (SPLIT)
- Pumping and reverse-dynamics effects (VCA with negative ratios)

Controls

Drag a knob up or down to turn it (hold Shift for fine moves), use the mouse wheel, or double-click to reset it. The value shows under the knob.

The preset row

PRESETS	Opens the list of factory presets (clicking the display does the same). Choosing one sets the compressor and side-chain controls; DRIFT, LEVEL, the EQ and the IN / OUT controls are left as they are. The last entry, Reset all to defaults, resets the compressor, DRIFT, LEVEL and IN / OUT.
Left / right arrows	Step to the previous or next preset.
Display	Shows the preset number and name.

The graph

SIGNAL	Lights when there is signal at the input.
INPUT	The input level in dB.
Transfer curve	Shows how input level (along the bottom, -60 to 0 dB) becomes output level for the current threshold, ratio and knee. The small triangle under the graph marks the threshold; the red square rides on the curve with the live level, and the arrow on the right shows the output level after MAKEUP.
Gain reduction meter	Lights from the right, 0 to -24 dB.

Colour swatch	Picks the faceplate colour: GREEN, BLUE or one of the GFX themes.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
HW / FLAT	Switches between the hardware-style faceplate and the flat look.

COMPRESSOR

THRESHOLD	The level where compression starts, -50 to 0 dB.
RATIO	How strongly the signal is compressed once it is over the threshold: from 1:1 up to INF:1 (limiting). In VCA only, the ratio carries on past INF into negative ratios, down to -1:1: the louder the input, the quieter the output - for pumping and swelling effects. The other voicings stop at INF:1.
KNEE	How gradually the compression eases in around the threshold, 0 - 18 dB (6 dB by default). RIDE always uses at least 10 dB and OPTO at least 12 dB.
MAKEUP	Gain after compression, 0 to +24 dB.
MIX	Parallel compression: 100% is fully compressed, lower settings blend the dry signal back in.

Timing and voicing

ATTACK	How fast the compressor reacts, 0.1 - 100 ms. Shows AUTO when AUTO is on and OPTO in the OPTO voicing.
RELEASE	How fast it lets go, 20 - 2000 ms. Shows AUTO or OPTO in the same way.

AUTO	Programme-dependent timing (on by default): the attack is set by the voicing and the release lengthens the longer the compressor keeps working, so sustained passages do not pump. Always on in OPTO.
VCA	The punchy loudness-sensing (RMS) compressor. It adds a touch of even harmonics under heavy reduction, and is the only voicing with negative ratios.
BUS	Peak-sensing, smooth and low in colour, for glue on groups and mixes. Always linked.
RIDE	A transparent 'easy ride' leveller with a wide knee and adaptive timing.
OPTO	An optical cell: a soft attack and a two-stage release that lets go quickly, then slowly. ATTACK and RELEASE are not used.
LINK	On (default): both sides get the same gain, so the stereo image stays still. Off: each side is compressed on its own. Always on in BUS.

SIDE-CHAIN

SC HPF	The compressor ignores the lows below the frequency (a 12 dB/oct high-pass on what it listens to), but the whole signal is compressed. Use it so the bass does not make everything pump. The label reads DETECTOR HIGH-PASS.
SPLIT	Only the lows below the frequency are compressed; everything above passes untouched. The split is a 24 dB/oct crossover that sums back flat when nothing is being compressed. The label reads LOWS ONLY COMPRESSED.
Frequency knob	Where the high-pass or split sits, 20 - 2000 Hz (80 Hz by default). Fully left is OFF.

IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB. Double-click to reset it.
GUARD (click TRIM)	A single click on the TRIM knob switches GUARD on or off; the small light beside the knob shows it (dark = off, red = on, bright = working). GUARD is on by default. It gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources.
SAT	Band saturation on the way out, 0 - 100%: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.

The right-hand column

DRIFT	A slow random wander, 0 - 100% (50% is the standard amount, 100% twice that, 0% none). The threshold breathes very slightly, the attack and release times move a little, and the EQ bands' frequencies and gains wander very slightly. The wander also gets a little faster as you turn it up.
Meter	Output level, left and right.
LEVEL	The output level, -24 to +12 dB.

EQ

Four bands in a row, each with a coloured gain knob and its FREQ knob beside it, between a low cut and a high cut. The bell bands get slightly narrower as you boost or cut further.

LO CUT	A 12 dB/oct high-pass filter, 20 - 1000 Hz. Fully left is OFF.
Gain knobs	Every band's gain runs from -16 to +16 dB. Below -12 dB the cut gets much steeper, reaching -40 dB at -16 - a deep notch on a bell band.

LOW / FREQ	Gain -16 to +16 dB, frequency 20 - 500 Hz.
LO MID / FREQ	A bell: gain -16 to +16 dB, frequency 80 Hz - 2 kHz.
HI MID / FREQ	A bell: gain -16 to +16 dB, frequency 400 Hz - 8 kHz.
HIGH / FREQ	Gain -16 to +16 dB, frequency 1.5 - 20 kHz.
HI CUT	A 12 dB/oct low-pass filter, 2 - 20 kHz. Fully right is OFF.
SHELF: LOW / HIGH	Turn the LOW and HIGH bands into shelves (lit, the default) or bells (off).
EQ FIRST / EQ AFTER	Puts the EQ before the compressor (it changes what the compressor hears and reacts to) or after it (it shapes the compressed sound). The panel swaps the two units so the first one is on top. The key's light shows while it is held.

Presets

Vocal Ride	RIDE at 3:1 with auto timing, a 12 dB knee, 3 dB make-up and a 100 Hz SC HPF.
Drum Smash	VCA at about 6:1, a 3 ms attack and 80 ms release, a near-hard knee and 6 dB make-up.
Bass Glue	VCA at 4:1 with auto timing, compressing only the lows below 120 Hz (SPLIT).
Bus Glue	BUS at 2:1, 30 ms attack, 300 ms release, 2 dB make-up and a 90 Hz SC HPF.
Negative Pump	VCA past INF at -2:1, for pumping and reverse-dynamics effects.
Gentle Level	RIDE at 2:1 with a very wide 15 dB knee and a slow auto release.
Parallel Crush	VCA at 10:1 with a hard knee and very fast timing, 10 dB make-up, blended in at 40%.

Mastering Touch	BUS at 1.5:1 with auto timing and a 9 dB knee, just 1 dB of make-up.
Opto Vocal	OPTO at 4:1 with a 12 dB knee, 4 dB make-up and an 80 Hz SC HPF.
Opto Lows Only	OPTO at 5:1 compressing only the lows below 160 Hz (SPLIT).
Reset all to defaults	The compressor, side-chain, DRIFT, LEVEL and IN / OUT back to their starting values. The EQ is left as it is.

Tips

- Watch the red square on the transfer curve: it shows exactly how far over the threshold the music sits.
- EQ FIRST with a LO CUT or a low cut on LOW makes the compressor react less to the bass; EQ AFTER keeps the compression the same and only changes the tone.
- For parallel compression, set heavy compression first, then bring MIX down until the transients come back.
- SPLIT on a mix bus at 100 - 150 Hz steadies the low end without touching the vocal or the cymbals.

File: gfx-compressor-q.jsfx